SCMS School of Engineering and Technology

Department of Computer Science and Engineering

Project Planning & Proposal Form

CSD 415 PROJECT PHASE I -BATCH 2023-2027

Project Title: AI-Powered Cursor Control for Instant Assistance

Group No: 01

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1. Purpose of Project

Modern users frequently switch between applications to obtain AI assistance by copying text, taking screenshots, or manually describing what is displayed on the screen. This fragments workflows and reduces productivity, since existing AI assistants provide limited contextual awareness and cannot actively assist users while they interact with websites, text, or desktop applications.

This project aims to build a fully software-based, cross-platform desktop overlay application that lets a user hold a hotkey, point to any screen element, and instantly receive AI-powered contextual assistance. The assistant captures the relevant screen region, understands it using a Vision Large Language Model (Vision-LLM), and displays the response in a lightweight floating widget near the cursor – without requiring the user to leave their current application.

Beyond these fixed capabilities, the assistant also functions as a self-tooling agent: a user can simply prompt it to build a new feature they need, and the agent will autonomously plan, generate, and integrate that capability into the application. Once created, the new feature is saved permanently as part of that user's personal installation, so the assistant's functionality keeps growing and adapting around each individual user over time.

2. Project Outline

The system consists of five modules:

- **Desktop Overlay & Input Capture** – uses global keyboard and mouse hooks to let the user trigger the assistant from anywhere on the system and capture a cursor-region screenshot.
- **Screenshot-based Contextual AI** – sends the captured region to a Vision-LLM to answer queries or explain on-screen content such as code, charts, or webpages.
- **Universal AI Text Assistant** – provides grammar correction, rewriting, summarization, explanation, translation, and word/phrase meaning for any selected or pasted text.
- **Website Trust/Spam Detection** – analyses a hovered link or domain and flags it as genuine or potentially unsafe.
- **Prompt-It – Self-Tooling Feature Agent** – lets a user describe a new capability they want in plain language; the agent autonomously designs, builds, and integrates that feature into the assistant, then permanently retains it as part of that user's personal toolset for future use.

3. Project outcomes/key deliverables at the end of VIII Semester

- Journal Publication ☐
- Conference Publication ☐
- Patent Publication ☐
- APP / Web Hosting ☑
- Hardware Product ☐
- Project Funding ☐
- Others (specify details) ☐

4. Resource Allocation

4.1 Proposed budget

Estimated total budget: Rs. 500 – Rs. 5000. The project is fully software-based and requires no dedicated hardware; the budget covers AI API usage credits and incidental costs only.

4.2 Technology Used

- Application Shell: Electron.
- Frontend: React, TypeScript, Tailwind CSS and Framer Motion.
- Build Tool: Vite.
- System Integration: Node.js with uiohook (global keyboard and mouse hooks) and Electron nativeImage.
- Local Storage and Security: electron-store with safeStorage for encrypted API keys.
- AI Providers: OpenAI, Google Gemini, Anthropic and OpenRouter (Vision-LLM APIs).

4.3 Other resources (Kit, Sensors, Cloud subscription, etc)

- No dedicated hardware or sensor kits required – a standard laptop or desktop is sufficient.
- Test machines covering Windows, macOS and Linux for cross-platform packaging.
- Free or trial-tier API credits from the AI providers listed above.
- GitHub repository for version control and release hosting.

5. Timeline with Milestones:

No.	TASK NAME	Jul-26		Aug-26		Sep-26		Oct-26		Nov-26		Dec-26		Jan-27	
		W1-W2	W3-W4	W1-W2	W3-W4	W1-W2	W3-W4	W1-W2	W3-W4	W1-W2	W3-W4	W1-W2	W3-W4	W1-W2	W3-W4
1	Requirement Analysis & Research														
2	Desktop Overlay & Input Capture														
3	Screenshot Capture & Vision-LLM Integration														
4	Prompt-It Self-Tooling Feature Agent Development														
5	Universal AI Text Assistant														
6	Website Trust / Spam Detection														
7	Settings & API Key Management, UI Polish														
8	Integration Testing & Debugging														
9	Packaging, Documentation & Demo Video														

Approved by,

Project Guide	Project Coordinator	HOD
Ms. Surya S G	**Ms. Surya S G**	**Dr. Manish T I**
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